

Authentication and Security Research

1. JSON Web Tokens (JWT)

An often used standard for stateless authentication in REST APIs is JWT.

Advantages

- No state and therefore scalable
- Works well with both web and mobile clients
- Supports role-based access control

JWT was selected to allow the least server overhead while securing the endpoints for both students and instructors. (Jones, Bradley and Sakimura, 2015).

2. Password Hashing Algorithms

Security research has pointed to the importance of hashing user passwords with the current methods that include Bcrypt and Argon2 as a part of the overall security strategy.

These procedures are shielding user-reported data from the onslaughts of rainbow-tables and brute-force. (OWASP, 2023).